CHE110 · Lecture 1 study guide

Built from the full lecture transcripts (L1P1, L1P2, L1P3), with your slides used only for context. Tags: TAUGHT IN CLASS said in the lecture · EXAM SIGNAL the lecturer said it is or will be tested · NOT YET TAUGHT on the slides but the lecturer said it comes in a later class.

What the lecturer said about the exam

1. **"Follow the class."** Questions come from what was taught in class, not from every line of the slides. The slides are extra detail "to clear your basics", not for line-by-line memorising.
2. **Maths will definitely appear** in the mid, the quiz and the final. The Rydberg-type problems are the core.
3. **Constants are given** in the question, with units: R_H, h and c. **Do not memorise them.**
4. **The formula is NOT given.** You must know it, **including the squares** (many students forget the square).
5. **No derivation needed.** "We don't need the formula derivation." Know what the formula gives you and where to apply it.
6. **Write the unit** with every answer. No unit conversion needed: leave λ in metres, ΔE in joules (no need for nm or Å unless asked).
7. **You may name n_1, n_2 as you like** (n_1/n_2, n_L/n_H, n_final/n_initial). What matters is placing the values correctly.
8. **MCQ, true/false and short questions** come from the conceptual points. "Written questions this big" (write the postulates, write the limitations) do **not** come from the models. You are not asked to memorise the postulates of Rutherford's and Bohr's models.
9. Do not memorise masses, charges and mass numbers of electron, proton and neutron.
10. Do not memorise which spectral region each series falls in.
11. Band spectrum is **not** studied.

Points the lecturer flagged as likely questions

Strong nuclear force holds neutrons + protons

Coulombic attraction holds electrons to nucleus

Rutherford vs Maxwell are contradictory (true/false)

Bohr: angular momentum is quantized

n = orbit number, not orbital

Which level is "initial"? (higher, n_2)

Find the initial level from λ (Balmer 656 nm)

Series order: Lyman, Balmer, Paschen, Brackett, Pfund

Lower limit n_1 of each series

Line vs continuous (white light = continuous)

E and λ are inversely related

Ionization energy is endothermic

Electrons absorb only exact energy packets

Nodes (explained in a later class)

What was taught, in order

#	Topic	Core takeaway
1	Recap of structure; forces in the atom	Atom is mostly empty. Nucleus is tiny but holds the mass. Strong nuclear force binds p and n. Neutron is the "peacemaker".
2	Rutherford	Discovered the nucleus. Solar-system comparison. Fails: a moving electron should lose energy (Maxwell) and fall in.
3	Bohr	Orbits. Fixed (quantized) energy. Angular momentum quantized: $mvr = nh/2\pi$. n = orbit number.
4	Energy levels, packets	Electron absorbs only the exact energy gap. Gaps between levels are empty, like stairs.
5	Emission	High → low releases $\Delta E = E_2 - E_1 = hc/\lambda$. We measure emitted, not absorbed, energy.
6	Rydberg formula	$1/\lambda = R_H(1/n_1^2 - 1/n_2^2)$. n_2 = initial = higher. Then $E = hc\cdot(1/\lambda)$.
7	Practice: ∞ → 1	$\lambda = 9.1\times10^{-8}$ m, $\Delta E = 2.18\times10^{-18}$ J.
8	Spectrum	Prism, VIBGYOR. Line vs continuous. Emission vs absorption. Lower λ means higher energy.
9	Five series	Fixed lower level per series. Closest level → smallest ΔE → largest λ. Balmer practice question.
10	Slide recap	Coulombic vs strong force, gold foil, ionization, semi-classical Bohr, line/continuous.

> **Not covered in this recording:** Dalton, isotopes/isobars and Thomson (the lecturer said "next class"), the max/min wavelength calculations (done in Lecture 2), nodes and de Broglie (Lecture 2). Since the quiz covers lectures 1–3 from the slides, Dalton, isotopes and Thomson are in the slides section; the rest are in the Lecture 2 guide.

1 · The nucleus and its forces

TAUGHT IN CLASS EXAM SIGNAL

Picture of the atom

- Neutrons and protons sit in the middle (the **nucleus**); electrons are around it.
- **Space:** most of the atom is **empty**. The nucleus takes up very little space; the electrons take up most of it. (That is what Rutherford found.)
- **Mass:** the nucleus is far heavier than the electrons.

Which force holds what?

What is held	Force
Neutrons and protons in the nucleus	**Strong nuclear force**
Electrons and the nucleus	**Coulombic (electrostatic) attraction** between negative and positive

> **The lecturer's words:** "In every semester a small question like this is given. With the help of which force do the neutron and proton stay together in the nucleus? … Strong nuclear force. Look, the force's name itself is nuclear force." A common wrong answer is magnetic force. Remember: *electron–nucleus = Coulombic; protons and neutrons = strong nuclear force.*

Why is a strong force needed?

- Protons carry **positive** charge. There is **no negative charge** inside the nucleus, so everything there is positive. Same charges **repel**.
- Many protons packed into a tiny space should fly apart. They don't, because the **strong nuclear force** binds them. The lecturer compared it to **glue** (gluon-like) that binds neutron and proton.
- The strong force works only over **very short distances** (a space too small to see). If you increase the distance between the particles, the strong force can no longer hold them.
- The nucleus does not break easily; ordinary atoms (H, N, O) stay intact. It breaks only if it is radioactive or hit by outside radiation. Very high atomic number → radioactive (a later topic).
- The **proton number defines the atom**. If the proton number changed, the atom would become a different element, so the atom does not let protons leave easily.

What are neutrons and protons made of?

Quarks (charged, positive and negative). For the **neutron**, the quark charges add and subtract to **zero**, so it is "zero charge". For the **proton**, they add up to **+1**.

Role of the neutron: the "peacemaker"

- Neutrons sit between protons and create a little distance, so repulsion is **reduced**.
- **n : p ratio of 1 : 1 → stable.** Example: carbon-12 (6 p, 6 n).

- **Too many neutrons** ("too many peacemakers become a problem") → the nucleus is **unstable and radioactive**. Example: carbon-14 (6 p, 8 n), used in **carbon dating**.
- Slight changes are fine: sodium has 11 p and 12 n. A ratio roughly between 1 and 1.5 causes no problem. Beyond that, the nucleus becomes radioactive.
- Excess protons *or* excess neutrons both lead to radioactivity ("if it is not balanced, it will be unstable").

$$\text{stable: n} \approx \text{p } (1:1 \text{ up to } \sim 1.5:1) \;\mid\; \text{unbalanced} \rightarrow \text{radioactive}$$

Notation

- An atom is written with its **mass number** (top left) and **atomic/proton number** (bottom left).
- **Mass number = protons + neutrons.** So neutrons = mass number − atomic number.

Atom	Mass no.	Protons	Neutrons	n : p
^{12}C	12	6	6	1 : 1 (stable)
^{14}C	14	6	8	1.33 : 1 (radioactive)
^{23}Na	23	11	12	1.09 : 1 (fine)

Electron mass

The electron's mass is about 10^{-31} kg. It is **negligible** compared with the neutron and proton, but **negligible is not the same as zero**. It has mass and it is countable. (You do not need to memorise the exact values.)

2 · Rutherford's model

TAUGHT IN CLASS EXAM SIGNAL

What Rutherford did and said

- **Discovered the nucleus**: a positively charged centre. He is "the father of the nucleus".
- He did **not** determine the proton and neutron numbers (how many of each).
- He compared the atom to the **solar system**: electrons revolve like planets.
- He said something about electron motion but gave **no mathematical explanation**.
- Gold-foil experiment: he passed alpha particles through the atom (foil). **Most went straight through** → the atom is mostly empty space. (Full result in the recap, section 10.)

The problem with Rutherford's model

> **Lecturer's argument.** An electron keeps **revolving**, so it has velocity. A moving body eventually **loses energy**. Planets are neutral. An electron is a **charged** particle moving around an opposite charge (the nucleus), so as it loses energy it comes closer to the nucleus and finally **falls in**. Then the atom would no longer exist.
>
> **Maxwell** said exactly this: a moving charged body loses energy and will fall. **Rutherford** implied it won't fall. Their statements are **contradictory**, and **Maxwell is correct** here.

> **True/false signal:** "Rutherford's and Maxwell's statements are opposite to each other" is **TRUE**. The lecturer noted that some students wrongly marked it false.

- The analogy itself is weak: planets are **neutral**, electrons are **charged**. "Their way and these guys' way will not be the same. Never will be."
- Many other things were not explained properly by the model.
- Its good aspect: the nucleus carries positive charge, and the atom is mostly empty space.

3 · Bohr's model

TAUGHT IN CLASS EXAM SIGNAL

Bohr "updated" Rutherford: the positive nucleus is fine, but the electron does not move randomly.

- Each electron has a **specific place**, called an **orbit**. The concept of orbit came first from Bohr's model. Orbits are **circular** (circumference $2\pi r$).
- Hydrogen has **one orbit** occupied (one electron). Atoms beyond H and He have more orbits.
- In a specific orbit, the electron has a **fixed velocity and fixed energy**. The word for fixed is **quantized** (a **discrete** level of energy).
- Every electron in the first orbit has the same energy; the second orbit has a different, fixed energy; the third has another. Each orbit carries a fixed energy and never a random one.
- Orbits are also called **shells** or **energy shells**.

Electron: wave and particle

The electron has mass (particle nature) and revolves like a wave (wave nature). So it has a **dual nature**, like light. It is never 50/50: if one nature is very high, the other is negligible. (This comes from **de Broglie's theory**, taught later.)

Angular momentum is quantized EXAM SIGNAL

$$m\,v\,r = n\,h\,/\,2\pi \quad n = 1, 2, 3, 4, \ldots$$

- The electron has mass *m*, velocity *v* and radius *r* from the centre, so it has **angular momentum** (mvr).
- Bohr said angular momentum increases in steps of **h/2π**. *n* says how many times: once, twice, three times… so *n* is a **natural number** (positive integer starting from 1).
- **MCQ / short-question signal:** "Which of the following did Bohr explain?" Answer: **angular momentum is quantized** (with respect to h/2π). He did **not** say "energy is quantized" directly; that came later. Quantizing one thing implies the rest (energy, velocity) get quantized.
- Bohr gave formulas but **couldn't explain beyond hydrogen**: for electrons in the 2nd, 3rd orbit he couldn't provide full mathematical expressions for energy and velocity.

> **n is the ORBIT number, not the orbital.** The values of n always denote the **orbit number or energy-shell number**. An *orbital* is a specific place that stays within an orbit (taught later). The lecturer: "I am not teaching orbital, I am teaching orbit."

Planck's constant, h

- **h = 6.626 × 10^{-34} J·s** (given in the exam).
- Role: a **parameter to compare the classical world with the quantum world**. Whenever you study anything about electrons in atoms, **h appears**, and everything is compared with it as the standard. It introduces you to the quantum world.

Why Bohr's model beats Rutherford's

It explains the electron's place mathematically: the place is not random, and a specific place means a fixed velocity and fixed energy.

> **"Semi-classical" model.** Bohr said angular momentum is quantized; the phrase "energy is quantized" came much later. His model is **semi-classical**: he couldn't fully move to quantum theory nor break out of classical theory. **But the concept of quantum theory started from him.** "Quantized" is only possible for tiny invisible objects like atoms; our everyday energy does not become quantized. (His full postulates and limitations are in the slides; a long written question on them is not expected.)

4 · Energy levels, packets and excitation

TAUGHT IN CLASS EXAM SIGNAL

The lecturer's example. Level 1 has energy **20 J** and level 2 has **50 J**. To go from level 1 to level 2 you need **30 J**. Photon energy packets of **10, 20, 25, 30 and 40 J** are sent. Which does the electron absorb? Try it:

Level 1 = 20 J, level 2 = 50 J. Needed = 30 J. Only the 30 J packet is absorbed.

- Only **30 J** is absorbed. The electron **never absorbs random energy**. It receives **exactly** the amount that pushes it to the higher level in one go.
- You cannot make 30 J by combining 10 J + 20 J: it won't "sit taking 10 J, waiting for 20 J". Packets are not added up.
- These energy packets are **photons**.
- **Quantized energy** = fixed energy carried, and fixed energy received.

Empty spaces between levels, and nodes

- Like **stairs**: you stand on one step or another, never in the gap between. You cannot climb halfway.
- Electrons are never found in these in-between spaces. The electron exists as a **cloud** with non-uniform density. At the edge of the cloud the probability of finding the electron is very low, essentially **zero**. Such places are called **nodes**.
- The lecturer said **questions will come from nodes**, but nodes will be explained properly in a later class. For now: **node = place where the probability of finding an electron is (close to) zero.**

Excited state and coming back

- An electron that receives the right energy goes to a higher level: the **excited level**.
- Nobody stays excited for long, so the electron comes back to a lower level. It must **release the extra energy**: from 50 J to 20 J it releases **30 J**.
- For hydrogen's single electron, only the first shell is in use normally; the other shells exist but matter only when the electron is excited.

5 · Emission and ΔE = hc/λ

TAUGHT IN CLASS EXAM SIGNAL

$$\Delta E = E_2 - E_1 = hc / \lambda$$

- **h** = Planck's constant; **c** = speed of light; **λ** = wavelength of the radiation.
- This gives the **energy difference**. (The lecturer noted this does not itself explain quantization; the formula for quantized energy is different.)
- Equivalent form: **ΔE = hc · (1/λ)**. If you already have 1/λ from the Rydberg formula, multiply by hc to get ΔE directly.

Why we measure differences, and why emission

- We always talk in **differences**: temperature "15° higher", "2° higher than yesterday". Energy is the same: compare before and after.
- The **absorbed** energy can't be seen or directly measured. Analogy: a sponge soaks up water. You can only measure how much water it held when you **squeeze it out (release)**.
- What we measure is the **emitted** energy (high → low). That is why we see electromagnetic radiation: these are all **emission rays**.
- The emitted radiation can be anywhere in the electromagnetic spectrum: radio, microwave, infrared, UV, visible. If it is visible, we see it.
- Absorbed wavelength can be found with a **separate instrument**, but the standard formulas in this course are for **emitted light**.

6 · The Rydberg formula

TAUGHT IN CLASS EXAM SIGNAL: maths in mid, quiz and final

$$1/\lambda = R_H \left(1/{n_1}^2 - 1/{n_2}^2 \right)$$

- R_H = Rydberg constant = **$1.097 \times 10^7\ m^{-1}$** (given in the exam, with units).
- **n_1 = lower level** (final level). **n_2 = higher level** (initial level).
- **Don't forget the squares!** They came from the derivation; you don't need the derivation, but you do need the squares.
- The formula gives the **λ of emitted light**.

> **Why n_2 (the higher level) is "initial".** The transition (emission) **starts** at the higher level and **ends** at the lower. So **initial = higher = n_2**, **final = lower = n_1**. "Find the initial level" means find **n_2**.
> The 1 and 2 are only serial labels: you can write them as n_lower / n_higher or n_final / n_initial. The lecturer said the naming is up to you; what matters is placing the values correctly.

Why the larger n goes in the second term. The start is the larger value, and mathematically the larger value has to be kept later, otherwise the result becomes negative. If your 1/λ comes out negative, you swapped n_1 and n_2.

Reading a transition

"The transition is 4 to 1" means the electron goes **from n = 4 to n = 1**. The first number is the **initial (higher)**, the second is the **final (lower)**.

Z and hydrogen

The general formula has a **Z** (atomic number). For hydrogen Z = 1, so it is not needed. The lecturer: questions are always set for **hydrogen**, so ignore Z.

Procedure (use every time)

1. Identify n_2 (initial, higher) and n_1 (final, lower).
2. Compute $1/\lambda = R_H(1/n_1^2 - 1/n_2^2)$. ($1/\infty = 0$.)
3. Invert: $\lambda = 1/(1/\lambda)$. Unit: m.
4. Energy: $\Delta E = hc \times (1/\lambda)$ in J.
5. Write the unit with the answer.

Common slips: forgetting the square · swapping n_1/n_2 (negative answer) · giving $1/\lambda$ instead of λ · forgetting the unit.

The lecturer's practice problem

The transition is from **n = ∞ to n = 1**. Find λ and ΔE. ($R_H = 1.097\times10^7$ m^{-1}, c = 3×10^8 m/s, h = 6.626×10^{-34} J·s.)
$1/\lambda = R_H(1/1^2 - 1/\infty^2) = R_H(1 - 0) = 1.097\times10^7$ m^{-1} → **$\lambda = 9.1\times10^{-8}$ m**
$\Delta E = hc/\lambda = (6.626\times10^{-34})(3\times10^8)(1.097\times10^7)$ = **2.18×10^{-18} J**
"1/infinity is zero." No unit conversion needed: leave λ in m.

7 · Energy rises with n; energy and wavelength

TAUGHT IN CLASS

Why does a farther level have more energy?

- Levels farther from the nucleus have **higher energy values**: $E_1 < E_2 < E_3$ … and $n = \infty$ has the most.
- The lecturer's reasoning (not about attraction!): to keep an electron far away, it has to be **carrying more energy**, "otherwise it won't be able to stay". The **value of that level itself is higher**. As n increases, velocity and radius also change, and the energy value rises.
- Don't confuse the two ideas: a far level has a higher **energy** but a **weaker attraction** to the nucleus. Higher energy does not mean higher attraction.
- It is **easier to ionize** an electron in a far level (weaker attraction), even though the level's energy is higher.
- No electrons have been seen beyond about the 7th shell for the 118 known elements.

> **Follow-up in Lecture 2.** At the start of Lecture 2 the lecturer, answering a student, said the first-orbit electron has the "highest" energy and that energy "decreases" going higher. She was talking about **binding energy** (how strongly the electron is held / the energy needed to free it), which is largest at n = 1 and falls with n. The **level energy** $E = -13.6\ \text{eV}/n^2$ still rises toward 0 as n grows. If a question says "ionization / energy needed to remove", it falls with n; if it says "energy of the orbit/level", it rises with n. Details in the Lecture 2 guide.

Energy and wavelength are inversely related

From $\Delta E = hc/\lambda$:

- **Large ΔE → small λ** (higher frequency, more vibration).
- **Small ΔE → large λ** (low frequency, slow waves).
- Biggest gap: n = ∞ to n = 1 → smallest λ. With n_1 fixed, **$n_2 = \infty$ gives the smallest λ**.
- Wavelength = distance from one crest to the next (or trough to trough). Lower wavelength → higher energy → higher frequency.

8 · Spectrum

TAUGHT IN CLASS MCQ / TRUE-FALSE SIGNAL

- White light through a **prism** gives **seven colours**: VIBGYOR (violet, indigo, blue, green, yellow, orange, red). All seven are inside white light. These sit in the **visible** range of the electromagnetic spectrum.
- Besides visible, radiation includes radio, microwave, infrared, UV and more; each has its own wavelength.
- A **spectrum** is understood in a broad sense, but also by specific lines.

Type	Meaning	Example
Line spectrum	A **specific wavelength** (a single colour per line). Lines are separate.	Atomic emission of H
Continuous spectrum	A **series** of wavelengths, more than one, together without gaps.	**White light**; red to violet
Band spectrum	Not studied.	—

> **Exam signal:** "White light: line or continuous?" → **Continuous**. White light has all colours, not one.

	Emission spectrum	Absorption spectrum
Origin	Light emitted (high → low)	Light absorbed by the substance
What you see	**Bright lines**, best seen on a **dark background**	**Dark** (no colour or radiation seen at those wavelengths); you place black on a white/coloured background

The lecturer said the background detail was only to help you understand ("these don't come in questions"). The conceptual statements (line vs continuous, emission bright vs absorption dark) can appear as true/false or MCQ.

- All the hydrogen series (Lyman, Balmer…) are **emission** series, so the transition is **high** → **low**.

9 · The five series

TAUGHT IN CLASS EXAM SIGNAL: know the order

> **Know the sequence** (the question will NOT give you n_1): **Lyman, Balmer, Paschen, Brackett, Pfund** = lower limit **n_1 = 1, 2, 3, 4, 5**.
> Mnemonic: **L**ong **B**efore **P**eople **B**ecame **P**unctual.

Series	Lower (final) level n_1	Electrons come from n_2 =
Lyman	**1**	2, 3, 4, … ∞
Balmer	**2**	3, 4, 5, … ∞
Paschen	**3**	4, 5, 6, … ∞
Brackett	**4**	5, 6, … ∞
Pfund	**5**	6, 7, … ∞

Why does a series have a lower limit?

In the lecturer's words: an electron sent to a high level can fall to *any* lower level (10 → 9, 9 → 7, 7 → 2…), which is random. Balmer instead **sent electrons up and fixed the lower level at 2**, so every transition from a higher level ends at 2 (3→2, 4→2, … ∞→2). Paschen fixed level 3; Lyman fixed level 1. Each series has a **lower limit** and "didn't work randomly". Each transition releases energy, so each fixed level gives a spectrum series.

Maximum and minimum wavelength (idea taught here; the maths was done in Lecture 2)

- Within a series, the transition from the **closest level** (3→2 in Balmer; 2→1 in Lyman) has the **smallest energy difference → maximum λ**.
- The transition from **infinity** has the **largest** energy difference → **minimum λ**.

Class question: find the initial level

The wavelength of a line in the **Balmer series** is 6.56×10^{-7} m. Find the initial energy level.
The initial level is $\mathbf{n_2}$ (the higher). Balmer → $n_1 = 2$.
$1/\lambda = 1/(6.56\times10^{-7}) = 1.524\times10^{6}\ m^{-1}$
$1.524\times10^{6} = 1.097\times10^{7}\ (1/4 - 1/n_2^{2}) \rightarrow 1/4 - 1/n_2^{2} = 0.1389 \rightarrow 1/n_2^{2} = 0.1111 \rightarrow \mathbf{n_2 = 3}$. The transition is 3 → 2.

The lecturer said you don't need to memorise the spectral region of each series. (In passing: Lyman is in the ultraviolet, Balmer in the visible, the rest infrared. The lecturer misspoke "infrared" for Lyman in class; it is not needed for the exam.)

10 · End-of-class recap (from the slides)

TAUGHT IN CLASS

- **Two facts to remember:** electrons and nucleus are attracted by **Coulombic attraction**; protons and neutrons stay together by the **strong nuclear force**.
- **Gold foil (slide picture):** the green balls are alpha particles. Three conclusions on the slide: most α pass through undeflected; some scatter at large angles; a very small number rebound by **180°**. The nucleus occupies a tiny space and is positive, and α is also positive, so like charges repel and the α goes back. Only 2–3 in a hundred or two hundred rebound in the picture. (The lecturer also described the deflected ones as being affected by the charges in the atom; the key idea is that α particles are positive.)
- **Electron stays in place** because the inward pull (the centre attracts) and the outward force from its motion balance each other.
- **Ionization:** to remove an electron you must give energy from outside, and something must also be ready to accept it. **Ionization energy is endothermic.** The orbit has a specific energy, so you must supply energy from outside before the electron leaves. Example: sodium gives an electron and becomes stable (octet) only because another atom takes it.
- **Bohr:** angular momentum quantized; "energy quantized" came much later; **semi-classical** model; quantum theory started from him.
- **Spectrum:** line = specific wavelengths, separate. Continuous = all wavelengths together (red to violet). White light is continuous.
- The lecturer joked that chemistry has exceptions (Kota Factory), because invisible things can behave differently in different situations.

> **Reading strategy from the lecturer:** read only as much as was taught in class. The slides have extra detail so you can clear your basics; understand them once, no line-by-line memorising.

Rydberg calculator

Use the procedure in section 6 to compute λ and ΔE (the interactive calculator is in the web version).

Worked problems

Constants (given in the exam): $R_H = 1.097\times10^{7}\ m^{-1}$, $h = 6.626\times10^{-34}$ J·s, $c = 3\times10^{8}$ m/s. Answers in metres and joules; no conversion needed.

1. (Class) ∞ → 1. $1/\lambda = R_H(1 - 0) \rightarrow \lambda = 9.1\times10^{-8}$ m; $\Delta E = hc/\lambda = 2.18\times10^{-18}$ J.

2. (Class) Balmer line, $\lambda = 6.56\times10^{-7}$ m. Find the initial level. $n_2 = 3$ (see section 9).

3. 3 → 2. $1/\lambda = R(1/4 - 1/9) = 5R/36 = 1.524\times10^{6}$ m^{-1} → $\lambda = 6.56\times10^{-7}$ m. $\Delta E = hc/\lambda = (6.626\times10^{-34})(3\times10^{8})/(6.56\times10^{-7}) = 3.03\times10^{-19}$ J.

4. 4 → 2 (Balmer). $1/\lambda = R(1/4 - 1/16) = 3R/16 = 2.057\times10^{6}$ m^{-1} → $\lambda = 4.86\times10^{-7}$ m. $\Delta E = 4.09\times10^{-19}$ J.

5. 4 → 1 (Lyman). $1/\lambda = R(1 - 1/16) = 15R/16 = 1.028\times10^{7}$ m^{-1} → $\lambda = 9.72\times10^{-8}$ m. $\Delta E = 2.04\times10^{-18}$ J.

6. 5 → 3 (Paschen). $1/\lambda = R(1/9 - 1/25) = 16R/225 = 7.80\times10^{5}$ m^{-1} → $\lambda = 1.282\times10^{-6}$ m. $\Delta E = 1.55\times10^{-19}$ J.

7. Smallest λ in Balmer ($n_2 = \infty$): $1/\lambda = R/4$ → $\lambda = 4/R = 3.65\times10^{-7}$ m. **Largest λ in Lyman** (2 → 1): $1/\lambda = 3R/4$ → $\lambda = 4/(3R) = 1.215\times10^{-7}$ m. Done on the board in Lecture 2, where it was called a common exam question. See the full table for all five series.

8. Packet problem (class). $E_1 = 20$ J, $E_2 = 50$ J → need 30 J. Only the 30 J packet is absorbed. On de-excitation 50 → 20 releases 30 J.

> **Check your answer:** bigger gap → smaller λ. ∞→1 gives about 9.1×10^{-8} m, the shortest for hydrogen. If your λ for 3→2 isn't about 6.56×10^{-7} m, recheck.

Flashcards

Question	Answer
Which force holds neutrons and protons together?	The strong nuclear force (like glue). Not magnetic.
Which force holds electrons to the nucleus?	Coulombic (electrostatic) attraction.
Why is a strong force needed in the nucleus?	Protons are all positive and repel; there is no negative charge inside. The strong force overcomes this over very short distances.
What are neutrons and protons made of?	Quarks. Neutron's quark charges sum to 0; proton's to +1.
Role of the neutron?	'Peacemaker': sits between protons, adds distance, reduces repulsion.
Stable n : p ratio?	About 1 : 1 (e.g. C-12: 6 p, 6 n); slightly above is OK (Na: 11 p, 12 n). Far off means radioactive.
Why is C-14 radioactive?	6 protons, 8 neutrons: too many neutrons, ratio not balanced. Used in carbon dating.
Mass number = ?	Protons + neutrons.
Rutherford discovered?	The nucleus (positive centre). He did not give the numbers of protons and neutrons.
Rutherford's analogy?	Solar system: electrons revolve like planets.
Rutherford's main problem?	A moving charged electron loses energy and would fall into the nucleus (Maxwell). Planets are neutral, electrons are charged.
True or false: Rutherford and Maxwell's statements are contradictory.	TRUE. Maxwell is correct.
What did Bohr introduce?	Orbits: specific places for electrons, with fixed (quantized) energy and velocity.
What did Bohr say is quantized?	Angular momentum: $mvr = nh/2\pi$. (Energy quantization came later.)
What is n in $mvr = nh/2\pi$?	Orbit number (energy shell), natural number 1, 2, 3… NOT orbital.
What is h and its role?	Planck's constant, 6.626×10^{-34} J·s. Parameter linking the classical and quantum worlds.

Why is Bohr's model 'semi-classical'?	It quantizes angular momentum but couldn't fully leave classical physics; but quantum theory started from it.
Does the electron have mass?	Yes ($\sim 10^{-31}$ kg). Negligible compared with p and n, but not zero.
Dual nature of the electron?	Particle (has mass) and wave. Never 50/50 (de Broglie, later).
Level 1 = 20 J, level 2 = 50 J. Which packet is absorbed?	30 J only. Never random energy; packets are not added (10 + 20 doesn't work).
What is a node?	A place where the probability of finding an electron is (close to) zero. Detailed later.
Excited electron comes back. What happens to the energy?	It releases the energy difference as radiation (50 → 20 J releases 30 J).
Energy-difference formula?	$\Delta E = E_2 - E_1 = hc/\lambda = hc \cdot (1/\lambda)$.
Why do we measure emitted, not absorbed, energy?	Absorbed energy can't be seen or measured directly (sponge analogy); released energy can.
Rydberg formula?	$1/\lambda = R_H(1/n_1^2 - 1/n_2^2)$. n_1 lower/final, n_2 higher/initial. Squares!
Which level is 'initial' for emission?	The higher level, n_2.
Is the formula given in the exam? The constants?	Formula: NO, learn it. Constants (R_H, h, c) with units: YES.
Is formula derivation needed?	No.
Do you convert λ to nm?	No. Write whatever unit comes out (m, J), but always write the unit.
Why can Z be ignored?	For hydrogen Z = 1; questions are about hydrogen.
λ and ΔE for ∞ → 1?	$\lambda = 9.1 \times 10^{-8}$ m; $\Delta E = 2.18 \times 10^{-18}$ J.
What does 'transition 4 to 1' mean?	Electron goes from n = 4 (initial) to n = 1 (final).
Energy vs n?	Energy of the level increases with n (n = ∞ highest). Higher energy is not stronger attraction.
Energy vs wavelength?	Inversely related. High energy: small λ, high frequency.
Line spectrum vs continuous?	Line: specific wavelength(s), separate. Continuous: many wavelengths together. White light = continuous.
Emission vs absorption spectrum?	Emission: bright lines (dark background). Absorption: dark (no light at those wavelengths).

Five series and their n_1?	Lyman 1, Balmer 2, Paschen 3, Brackett 4, Pfund 5. The order is not given in the question.
Why does each series have a fixed lower limit?	Electrons are sent to higher levels and the lower level is fixed; all transitions end there (not random).
Which transition gives the max λ in a series? The min?	Max λ: from the nearest level (smallest ΔE). Min λ: from ∞ (biggest ΔE).
Is ionization energy endothermic or exothermic?	Endothermic: energy must be supplied from outside.
Why is $n_2 > n_1$ always?	Otherwise 1/λ would be negative; the start is the larger value.
The Lyman, Balmer… series are examples of?	Emission spectra (transition high → low).

Quiz & drills

1. With the help of which force do the neutron and proton stay together in the nucleus?
   A) Magnetic force B) Strong nuclear force C) Coulombic attraction D) Gravitational force
   Answer: B. Strong nuclear force. Coulombic attraction acts between electron and nucleus.
2. Electrons are held to the nucleus by
   A) Strong nuclear force B) Coulombic attraction C) Peacemaker neutrons D) Magnetic force
   Answer: B. Opposite charges attract: Coulombic (electrostatic) attraction.
3. True or false: Rutherford's and Maxwell's statements about the moving electron are opposite to each other.
   A) True B) False
   Answer: A. Rutherford implied the electron wouldn't fall; Maxwell said a moving charged body loses energy and will. Maxwell is right.
4. Why is the neutron called a peacemaker?
   A) It carries positive charge B) It reduces repulsion between protons C) It orbits the nucleus D) It attracts electrons
   Answer: B. It sits between protons and creates distance, reducing repulsion.
5. Which nucleus is stable?
   A) Carbon with 6 p and 6 n B) Carbon with 6 p and 8 n C) Any with many neutrons D) None
   Answer: A. 1 : 1 is stable. 6 p / 8 n (C-14) is radioactive.
6. Mass number equals
   A) Protons B) Neutrons C) Protons + neutrons D) Electrons + protons
   Answer: C. Mass number = protons + neutrons.

7. Rutherford discovered

   A) Proton B) Neutron C) Nucleus D) Quantized orbits

   Answer: C. He is the 'father of the nucleus'; he did not give p and n numbers.

8. What is the main problem with Rutherford's model?

   A) Nucleus is negative B) A moving charged electron should lose energy and fall in C) Electrons are neutral D) Atom is full, not empty

   Answer: B. Maxwell's argument.

9. Bohr said which quantity is quantized?

   A) Mass B) Angular momentum C) Charge D) Temperature

   Answer: B. Angular momentum = nh/2π. Energy quantization came later.

10. In mvr = nh/2π, n is

    A) Orbit number (1, 2, 3…) B) Orbital type C) Number of neutrons D) Spin

    Answer: A. n is the orbit/energy-shell number, not the orbital.

11. Planck's constant is

    A) $1.097\times10^{7}\ m^{-1}$ B) 6.626×10^{-34} J·s C) 3×10^{8} m/s D) 9.109×10^{-31} kg

    Answer: B. R_H, c and the electron mass are the others.

12. Bohr's model is called

    A) Quantum B) Semi-classical C) Classical D) Wave-mechanical

    Answer: B. Quantized angular momentum, but still classical orbits.

13. Level 1 = 20 J, level 2 = 50 J. A packet of which energy raises the electron?

    A) 10 J B) 25 J C) 30 J D) 40 J

    Answer: C. The gap is 30 J. Random energies are never absorbed.

14. Can 10 J and 20 J packets together excite the electron across a 30 J gap?

    A) Yes B) No

    Answer: B. Packets are not added up. It needs exactly 30 J in one go.

15. Electron moves from the 50 J level to the 20 J level. It

    A) Absorbs 30 J B) Releases 30 J C) Releases 70 J D) Does nothing

    Answer: B. High to low releases the difference.

16. In the Rydberg formula, the initial level of an emission is

    A) n_1 (lower) B) n_2 (higher) C) Always 1 D) Always ∞

    Answer: B. The transition starts at the higher level.

17. The Rydberg constant R_H is about

    A) 6.626×10^{-34} B) 3×10^{8} C) $1.097\times10^{7}\ m^{-1}$ D) 9.1×10^{-31}

    Answer: C. Constants are given in the exam, but know which is which.

18. Which is true about the exam?

    A) Formulas are given; constants must be memorised B) Constants are given; the formula must be known C) Both are given D) Neither is given

    Answer: B. The formula is not given; constants come with units.

19. For $\infty \rightarrow 1$ in hydrogen, ΔE is about

A) 2.18×10^{-18} J B) 2.18×10^{-18} m C) 9.1×10^{-8} J D) 6.56×10^{-7} J

Answer: A. $hcR_H = 2.18\times10^{-18}$ J. ($\lambda = 9.1\times10^{-8}$ m.)

20. A Balmer line has $\lambda = 6.56\times10^{-7}$ m. The initial level is

A) 2 B) 3 C) 4 D) 5

Answer: B. $1/4 - 1/n_2^2 = 0.1389 \rightarrow n_2 = 3$.

21. The Balmer series has lower level n_1 =

A) 1 B) 2 C) 3 D) 4

Answer: B. Lyman 1, Balmer 2, Paschen 3, Brackett 4, Pfund 5.

22. Which is the correct order of the series?

A) Lyman, Balmer, Paschen, Brackett, Pfund B) Balmer, Lyman, Paschen, Pfund, Brackett C) Paschen, Lyman, Balmer, Brackett, Pfund D) Pfund, Brackett, Paschen, Balmer, Lyman

Answer: A. n_1 = 1 to 5.

23. Within a series the maximum wavelength comes from

A) $n_2 = \infty$ B) The nearest higher level C) The lowest level D) Random levels

Answer: B. Smallest ΔE means largest λ (e.g. 3→2 in Balmer).

24. The minimum wavelength in a series comes from

A) $n_2 = n_1 + 1$ B) $n_2 = \infty$ C) $n_2 = 2$ D) $n_1 = \infty$

Answer: B. Biggest ΔE means smallest λ.

25. Energy and wavelength are

A) Directly proportional B) Inversely related C) Unrelated D) Equal

Answer: B. $\Delta E = hc/\lambda$.

26. Energy of an orbit

A) Decreases as n increases B) Increases as n increases C) Is the same in all orbits D) Is random

Answer: B. Farther level, higher energy value ($n = \infty$ is highest).

27. White light gives a

A) Line spectrum B) Continuous spectrum C) Band spectrum only D) No spectrum

Answer: B. All seven colours / many wavelengths together.

28. A line spectrum shows

A) A specific wavelength per line B) All wavelengths C) Only absorption D) No colours

Answer: A. Specific, separate lines.

29. Emission spectra show

A) Dark lines B) Bright lines C) Only white D) No lines

Answer: B. Absorption is dark.

30. Ionization energy is

A) Exothermic B) Endothermic C) Zero D) Negative always

Answer: B. Energy must be supplied from outside.

31. The Lyman, Balmer… series are series of

A) Absorption spectra B) Emission spectra C) Band spectra D) Continuous spectra

Answer: B. Transitions go high → low.

32. A node is a region where

A) Electron density is highest B) The probability of finding the electron is ~zero C) The nucleus sits D) Photons are absorbed

Answer: B. Explained properly in a later class.

33. [Slides] Dalton's postulate that atoms of an element have identical mass is contradicted by

A) Isotopes B) Isobars only C) Electrons D) Molecules

Answer: A. Isotopes: same Z, different mass (^{1}H, ^{2}H, ^{3}H).

34. [Slides] Te (Z = 52) and I (Z = 53) both have mass number 127. They are

A) Isotopes B) Isobars C) Isotones D) Isomers

Answer: B. Same mass number, different atomic number.

35. [Slides] ^{14}C, ^{15}N and ^{16}O all have 8 neutrons. They are

A) Isotopes B) Isobars C) Isotones D) Ions

Answer: C. Same neutron number.

36. [Slides] Which law says a compound always has the same proportion of elements?

A) Conservation of mass B) Constant composition C) Boyle's law D) Avogadro's law

Answer: B. Law of constant composition (e.g. H_2O is always 2 H : 1 O).

37. [Slides] Dalton did not distinguish between

A) Atom and molecule B) Proton and neutron C) Cathode and anode D) Orbit and orbital

Answer: A. The molecule is the smallest part of a compound.

38. [Slides] In Thomson's cathode-ray experiment, the beam bends toward the

A) Negative plate B) Positive plate C) Both D) Neither

Answer: B. So the rays are negatively charged electrons.

39. [Slides] Thomson's model is called

A) Planetary model B) Plum-pudding model C) Nuclear model D) Semi-classical

Answer: B. Negative electrons embedded in a positive sphere.

40. [Slides] Which was a limitation of Thomson's model?

A) No nucleus B) Electrons have no charge C) It predicted line spectra correctly D) It used Planck's constant

Answer: A. It couldn't explain the gold-foil scattering.

41. [Slides] Rutherford's model predicts a continuous spectrum but atoms show

A) A continuous spectrum B) A line spectrum C) No spectrum D) A rainbow

Answer: B. This is one of its limitations.

42. [Slides] An incandescent bulb gives a

A) Line spectrum B) Continuous spectrum C) Absorption spectrum only D) Emission lines only

Answer: B. Like white light: all wavelengths.

43. [Slides] The lowest-energy orbit of an electron is called the

A) Excited state B) Ground state C) Node D) Orbital

Answer: B. Ground state.

44. Which statement about the electron's mass is right?

A) It is exactly zero B) It is negligible but not zero C) It is larger than a proton D) It equals a neutron's

Answer: B. About 10^{-31} kg; negligible vs p and n.

One-page cheat sheet

Forces

- Neutron + proton: **strong nuclear force**
- Electron + nucleus: **Coulombic attraction**
- Neutron = peacemaker; n:p ≈ 1:1 stable; too many n → radioactive (C-14)

Models

- Rutherford: nucleus (+), mostly empty, solar system. Fails: a moving charge loses energy (Maxwell ✓), falls in.
- Bohr: orbits, fixed (quantized) energy, **angular momentum quantized**: $mvr = nh/2\pi$, n = orbit no. (not orbital). Semi-classical.

Energy

- Absorb exactly ΔE, never random packets
- High → low = emission
- Energy ↑ with n
- $E \propto 1/\lambda$
- Ionization: endothermic

Formulas

- $mvr = nh/2\pi$
- $\Delta E = E_2 - E_1 = hc/\lambda = hc\cdot(1/\lambda)$
- $\mathbf{1/\lambda = R_H(1/n_1^2 - 1/n_2^2)}$, $n_2 > n_1$ (initial = n_2)

Given in exam: $R_H = 1.097\times10^7\ m^{-1}$, $h = 6.626\times10^{-34}$ J·s, $c = 3\times10^8$ m/s. **Not given:** the formula.

- Write units. No conversion needed.
- $1/\infty = 0$. Don't forget squares.

Series (n_1): Lyman 1 · Balmer 2 · Paschen 3 · Brackett 4 · Pfund 5.
Min λ: from ∞. Max λ: from the nearest level.
Spectrum: line = specific λ; continuous = many λ (white light); emission = bright lines; absorption = dark.

Slides deep-dive: Dalton, isotopes, Thomson and slide details

SLIDES (L1) These come from the Lecture 1 slides. The lecturer said Dalton and isotopes/isobars would be taught later and that the quiz covers lectures 1–3 from the slides, so learn them. The lecturer's advice still applies: understand once, don't memorise line by line.

Matter and atoms

- **Matter** = anything with mass that occupies space. It is a pure substance (element or compound) or a mixture.
- **Element:** made of only one kind of atom. **Compound:** two or more kinds of atoms joined in a **definite proportion**.
- **Atom:** extremely small particle of matter that retains its identity during chemical reactions. Everything (a table, a chair, you) is made of atoms.
- Particle data (slide; the lecturer said don't memorise the numbers): electron 1.602×10^{-19} C (negative), 9.109×10^{-31} kg (>1800× lighter than hydrogen); proton $+1.602\times10^{-19}$ C, 1.672×10^{-27} kg; neutron 0 C, 1.674×10^{-27} kg. The nucleus holds protons and neutrons (except hydrogen: one proton).
- Atoms have size of the order of 10^{-10} m. Protons and neutrons make up the nucleus and nearly all the mass; electrons take up almost all the volume.

Dalton's atomic theory (1766–1844)

Democritus first suggested the atom; Dalton placed it on a solid footing about two millennia later. Based on two laws:

- **Law of conservation of mass:** matter is neither created nor destroyed in a closed system (why we balance equations). Example: 10 g of wood burned in a sealed container gives 10 g of smoke, ash and gas.
- **Law of constant composition:** a pure compound always has the same proportion of the same elements. NaCl is always the same Na : Cl ratio; H_2O is always 2 H : 1 O, whether tap or river water.

1. All matter is composed of minute particles called atoms; atoms are **indivisible and indestructible** and retain identity in a reaction.
2. All atoms of a given element have the same properties, including mass.
3. Atoms of different elements have different properties and masses.
4. A compound is atoms of two or more elements chemically combined in **fixed proportions** (water: H : O = 2 : 1).
5. A chemical reaction is the **rearrangement** of atoms into new combinations. Only atoms take part.
6. Atoms are not created, destroyed or broken into smaller particles in a reaction.

Limitations of Dalton's theory

Dalton said	Why it fails
Atoms are indivisible	Atoms have protons, neutrons and electrons.
Atoms of the same element have the same mass	**Isotopes**: hydrogen has masses 1, 2 and 3.
Atoms of different elements have different masses	**Isobars**: Te (Z = 52) and I (Z = 53) both have mass 127.
Atom is the smallest part of both element and compound	The atom is the smallest part of an element that can exist freely; the **molecule** is the smallest part of a compound. Dalton didn't distinguish them.

Isotopes, isobars, isotones

Term	Same	Different	Example
Isotopes	Atomic (proton) number Z	Mass number A (neutrons)	^{1}H, ^{2}H, ^{3}H; ^{12}C, ^{14}C
Isobars	Mass number A	Z	^{127}Te and ^{127}I; ^{40}Ar, ^{40}K, ^{40}Ca
Isotones	Neutron number N	Z and A	^{14}C, ^{15}N, ^{16}O (all N = 8, as on the slide)

> **Memory trick.** iso**T**ope = same pro**T**ons (same element); iso**B**ar = same **B**ulk (mass number); iso**TONE** = same neutr**ONE**s. A = Z + N.

Isotopes differ in relative atomic mass but **not in chemical properties**; a radioactive form of an element is an isotope too.

Thomson's model (1897) and the electron

- **Cathode-ray tube:** two electrodes (negative cathode, positive anode) sealed in a glass tube with the air removed, connected to a high-voltage supply. The glass glows green. Rays leave the cathode, and the beam **bends away from the negative plate and toward the positive plate**, so cathode rays are **negatively charged particles (electrons)**, a constituent of all matter.
- Atoms are neutral, so Thomson proposed the **plum-pudding model**: negative electrons embedded in a diffuse sphere of positive charge. Postulate 1: positively charged sphere with electrons embedded. Postulate 2: atom is neutral because + and − charges are equal in magnitude.
- **Limitations:** could not explain how the positive charge holds the electrons or the stability of the atom; no nucleus; couldn't explain Rutherford's scattering experiment.

Rutherford: extra slide details

- 1911, gold foil (α-scattering). α particles come from a radioactive source at very high speed (about 180,000 miles per second on the slide).
- **Postulates (slide):** most of an atom is empty; most of the mass (99.95% or more) is in the tiny, positive nucleus; electrons move around the nucleus like planets (solar-system / planetary / nuclear model); number of electrons = positive charge of the nucleus so the atom is neutral; the inward electrostatic (centripetal) force is exactly balanced by the outward centrifugal force.
- Nuclei have diameters about 10^{-15} m; atoms about 10^{-10} m (so the atom is ~10^5 times wider).
- **Limitations (slide):** (1) By Maxwell, a rotating charged body must radiate energy continuously, so the electron would spiral into the nucleus. (2) Newton's laws apply to neutral bodies such as planets, not charged electrons; electrons repel one another while planets attract. (3) No idea about the shape of the orbits. (4) No explanation for many-electron atoms. (5) Continuous radiation would give a continuous (band) spectrum, but atoms give a **line spectrum**.

Bohr: extra slide details

- 1913. The first model with **quantum energy levels**; Nobel Prize in 1922.
- **Postulates (slide):** (1) Electrons revolve in special discrete orbits without radiating energy. (2) Each orbit is a shell / energy level with a specific amount of energy. (3) Absorbing energy to move from a lower to a higher level gives an **excited state**. (4) Radiating energy when moving from higher to lower. (5) Moving between orbits emits photons, producing characteristic absorption and emission spectra; since every element has its own signature, spectra identify materials. (6) Closer orbits (lower energy) are more stable; the lowest-energy orbit is the **ground state**.
- Quantization of angular momentum: $mvr = nh/2\pi$. Energy radiated when an electron goes from E_2 to E_1 is $E_2 - E_1 = h\nu$.
- **Limitations (slide):** only single-electron species (hydrogen-like); cannot explain the fine structure of spectral lines (implies sub-levels → new quantum numbers); cannot explain splitting of lines in magnetic or electric fields; electrons jumping between orbits isn't justified because of uncertainty in position and velocity; circular instead of elliptical path.

Spectrum definitions (slides)

- White light through a prism resolves into colours of different wavelengths. Newton (17th century) showed sunlight contains many colours.
- **Continuous spectrum:** contains all wavelengths (rainbow). **Line spectrum:** only certain colours / specific wavelengths.
- Slide Q: is an incandescent bulb's spectrum continuous or line? **Continuous**: it is like white light and disperses into all wavelengths without gaps.
- **Emission spectrum:** heated atoms emit light of certain wavelengths: **bright lines** (electrons jumping from excited to ground state). Each element has a unique bright-line spectrum.

- **Absorption spectrum:** white light through a substance leaves **dark lines** where wavelengths were absorbed.

> **Next:** Lecture 2 (de Broglie, nodes, quantum numbers, electron configuration) is in the Lecture 2 guide.